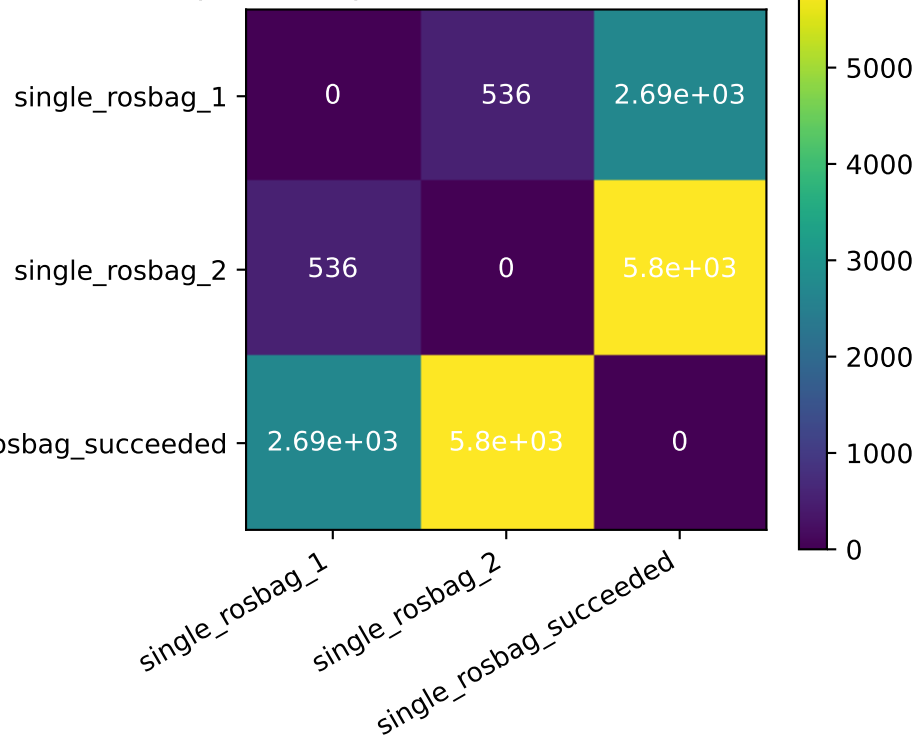
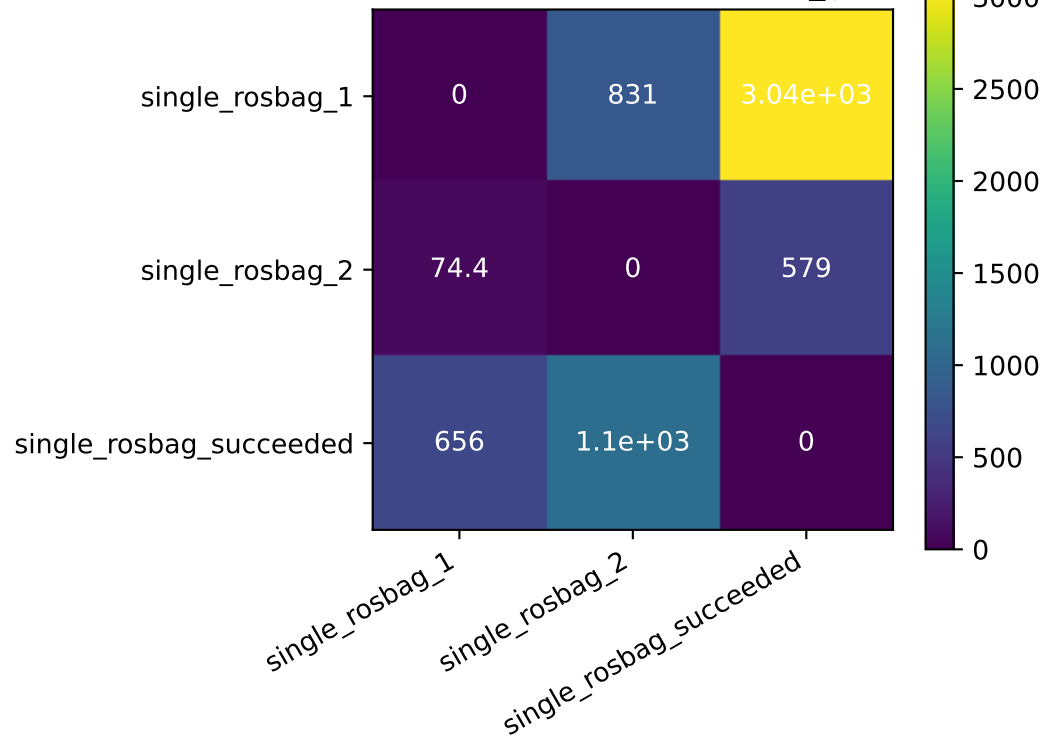


Three-bag scale-quotient consensus

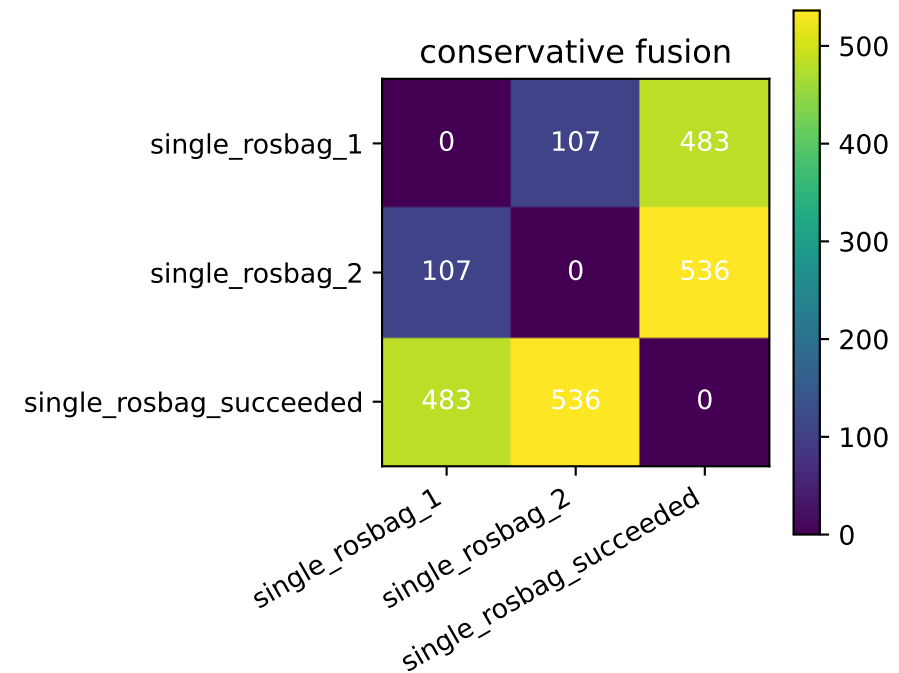
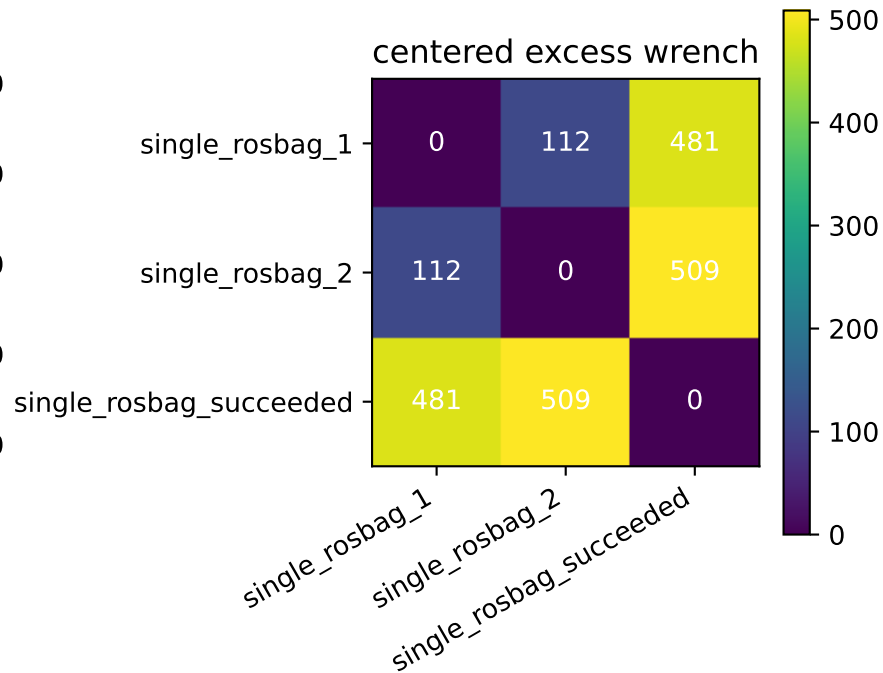
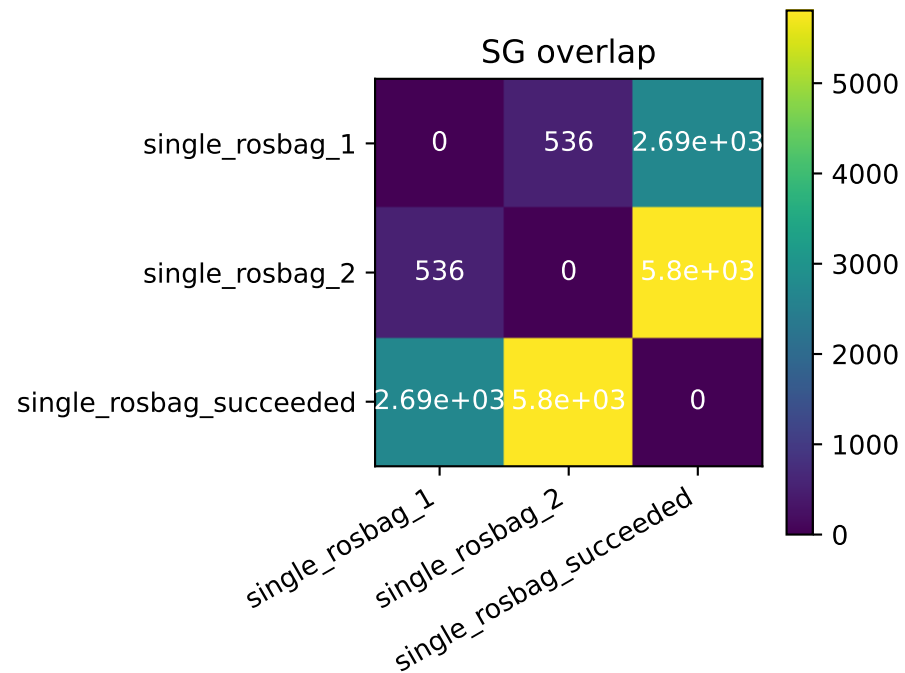
pairwise quotient distance: SG overlap



cross-evaluation Delta L_{ij}



Three-bag quotient covariance distances



Three-bag physical and cross-evaluation audit

Gauge-invariant physical quantities

single_rosbag_1

```
J/m = [[ 1.015805e-01 -1.228804e-04 -1.649156e-02]
[-1.228804e-04  7.735542e-02  9.522580e-06]
[-1.649156e-02  9.522580e-06  1.715480e-01]]
f/m = [0.334909 0.431263 0.376598 0.312238]
c      = [ 0.017628 -0.004117 -0.044376]
cross costs = [ 268.528887 1099.385419 3305.442753]
```

single_rosbag_2

```
J/m = [[ 0.073724  0.001825 -0.00043 ]
[ 0.001825  0.034796 -0.010867]
[-0.00043  -0.010867  0.105213]]
f/m = [0.381802 0.309009 0.34086  0.409455]
c      = [-0.026435  0.011339  0.009415]
cross costs = [187.059741 112.652483 691.905564]
```

single_rosbag_succeeded

```
J/m = [[ 0.231928 -0.005147  0.018047]
[-0.005147  0.10967  -0.003059]
[ 0.018047 -0.003059  0.128895]]
f/m = [0.444531 0.435005 0.229507 0.27868 ]
c      = [-0.019823 -0.055131 -0.005031]
cross costs = [ 684.664834 1130.166667  28.342745]
```

Three-bag conservative fusion diagnostic

Conservative quotient Gaussian product

fused precision rank: 13 / 13

unresolved fused direction count: 0

fused coordinate:

```
[ 2.1285991e-01  7.8968460e-01  3.8488878e+00  8.1690378e-01  
 -7.8671664e-01 -3.7700230e-01 -1.9722618e-02  2.4090674e-03  
 -1.8052524e-02 -6.7197909e-01 -7.2328219e-01 -8.2684129e-01  
 -6.1150447e-01]
```

fused covariance eigenvalues:

```
[7.4298512e-10 1.6959340e-09 2.1015561e-08 4.0245661e-08 1.6950625e-06  
 3.8751554e-06 1.3870109e-05 1.8676511e-05 5.2279796e-05 2.2787849e-04  
 2.5077936e-04 4.2557411e-04 6.3662777e-04]
```

bag-to-fused distance:

```
[457.8900383 917.5607312 812.6122114]
```

Interpretation: conservative local Gaussian uncertainty for fusion; not a calibrated generative covariance.